Line Maintenance Operational Dashboard

Live app: https://jfk-ops-live-dashboard-vercel-mirro.vercel.app/

Executive Summary

The Line Maintenance Operational Dashboard is a low-code, station-tailorable web application that brings daily roster planning, live operational visibility, weather awareness, and resource assignment into one execution surface. It was designed for practical line maintenance decision-making under time pressure, where fragmented systems and static reports often create delays and coordination gaps.

Core value: one operational board for engineers and managers to plan, execute, and rebalance workload in real time.

Problem Statement

- Daily roster files vary by station and format, making consistent interpretation difficult.
- Assignment coordination between certifiers and mechanics is often manual and time-sensitive.
- Operational context (weather, live movement, timeline congestion) is spread across multiple tools.
- Managers need fast visibility of peak workload windows to avoid resource bottlenecks.

Solution Design

1) Interpretative roster ingestion

A Daily Roster CSV upload module acts as the baseline input layer. The ingestion logic interprets roster content and transposes it into a structured operational board, including flight lines and assignment-ready records. This enables compatibility with varying station roster layouts and reduces setup friction.

2) Live execution board

- Flight list with status, A/C registration, A/C type, terminal/gate, ETA/STD, and notes.
- Editable assignment columns for certifier/mechanic with rapid operational updates.
- Column-level reset controls for quick board hygiene during dynamic changes.
- Station weather tile (current + day-part outlook) for tactical awareness.

3) Manager analytics view

- 24-hour Gantt timeline for flight overlap and sequencing visibility.
- Heat-map intensity overlay to identify peak workload periods.
- Supports staffing rebalance decisions and shift handover quality.

Enrichment & Data Resilience

The dashboard uses a layered enrichment architecture to reduce single-source dependency and maintain operational continuity.

- **Primary/secondary feed strategy:** modular provider routing with fallback handling.
- **Cache layer:** retains last-known useful values to reduce blank operational fields.
- **Manual override layer:** allows frontline correction where external feeds are incomplete.
- **Type normalization:** DOC8643-based aircraft type mapping to cleaner, standardized outputs.

Key Capabilities

- Rapid daily deployment from CSV input to actionable roster board.
- Resource assignment controls with practical reset and correction workflows.
- Real-time weather + operational context in one interface.
- Timeline and peak-demand visibility for management decisions.
- Low-code architecture, easy to tailor by station without high licensing overhead.

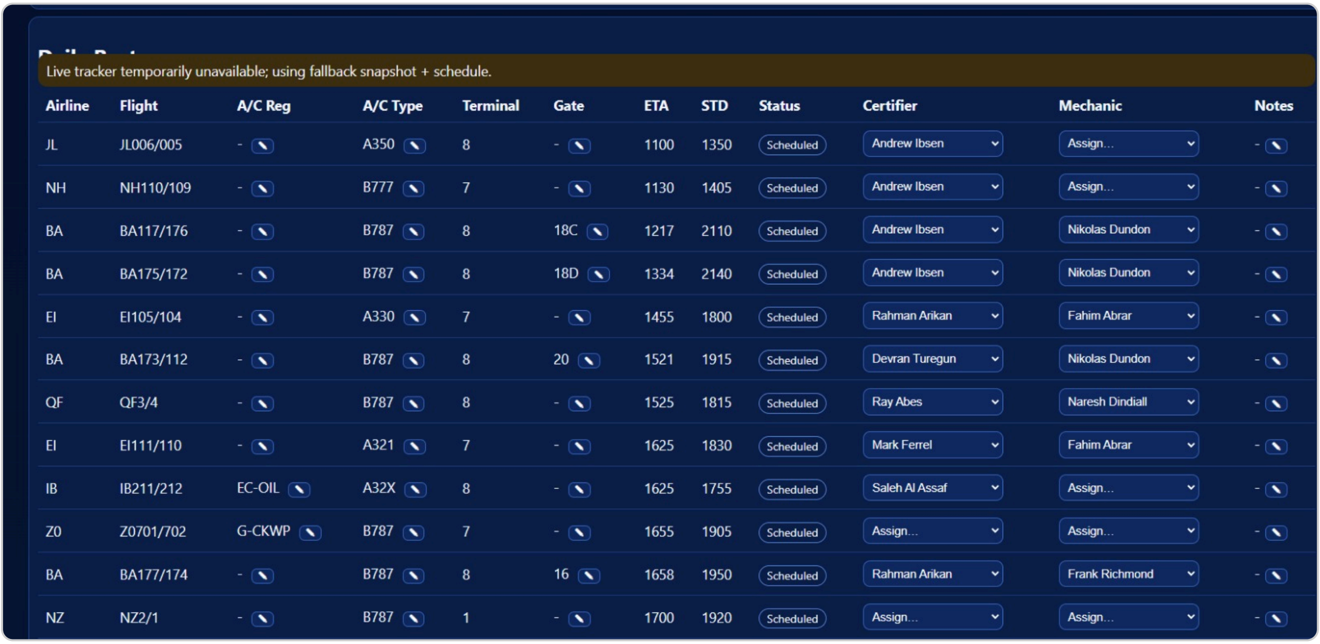
Practical Use Cases

- **Shift startup:** load roster and publish a consistent board view in minutes.
- **In-day disruption handling:** quickly reassign certifier/mechanic resources.
- **Handover quality:** preserve notes, assignment intent, and timeline clarity.
- **Peak-time planning:** use heat-map/Gantt to pre-position capability before congestion windows.
- **Multi-station scaling:** replicate core app while tailoring parser and assignment defaults locally.

Why this approach works

- **Operationally native:** built around line maintenance workflows, not generic project tooling.
- **Flexible input model:** adaptable to real-world roster variability.
- **Low-cost execution:** web-first, lightweight, and easy to maintain.
- **Human-in-the-loop by design:** automation accelerates decisions, while engineers/managers keep control.

Screenshots


Live tracker temporarily unavailable; using fallback snapshot + schedule.

Airline	Flight	A/C Reg	A/C Type	Terminal	Gate	ETA	STD	Status	Certifier	Mechanic	Notes
JL	JL006/005	-	A350	8	-	1100	1350	Scheduled	Andrew Ibsen	Assign...	-
NH	NH110/109	-	B777	7	-	1130	1405	Scheduled	Andrew Ibsen	Assign...	-
BA	BA117/176	-	B787	8	18C	1217	2110	Scheduled	Andrew Ibsen	Nikolas Dundon	-
BA	BA175/172	-	B787	8	18D	1334	2140	Scheduled	Andrew Ibsen	Nikolas Dundon	-
EI	EI105/104	-	A330	7	-	1455	1800	Scheduled	Rahman Arikan	Fahim Abrar	-
BA	BA173/112	-	B787	8	20	1521	1915	Scheduled	Devran Turegun	Nikolas Dundon	-
QF	QF3/4	-	B787	8	-	1525	1815	Scheduled	Ray Abes	Naresh Dindiall	-
EI	EI111/110	-	A321	7	-	1625	1830	Scheduled	Mark Ferrel	Fahim Abrar	-
IB	IB211/212	EC-OIL	A32X	8	-	1625	1755	Scheduled	Saleh Al Assaf	Assign...	-
Z0	Z0701/702	G-CKWP	B787	7	-	1655	1905	Scheduled	Assign...	Assign...	-
BA	BA177/174	-	B787	8	16	1658	1950	Scheduled	Rahman Arikan	Frank Richmond	-
NZ	NZ2/1	-	B787	1	-	1700	1920	Scheduled	Assign...	Assign...	-

Figure 1: Daily roster execution board. Editable assignment columns, operational status, and structured flight line view for shift control.

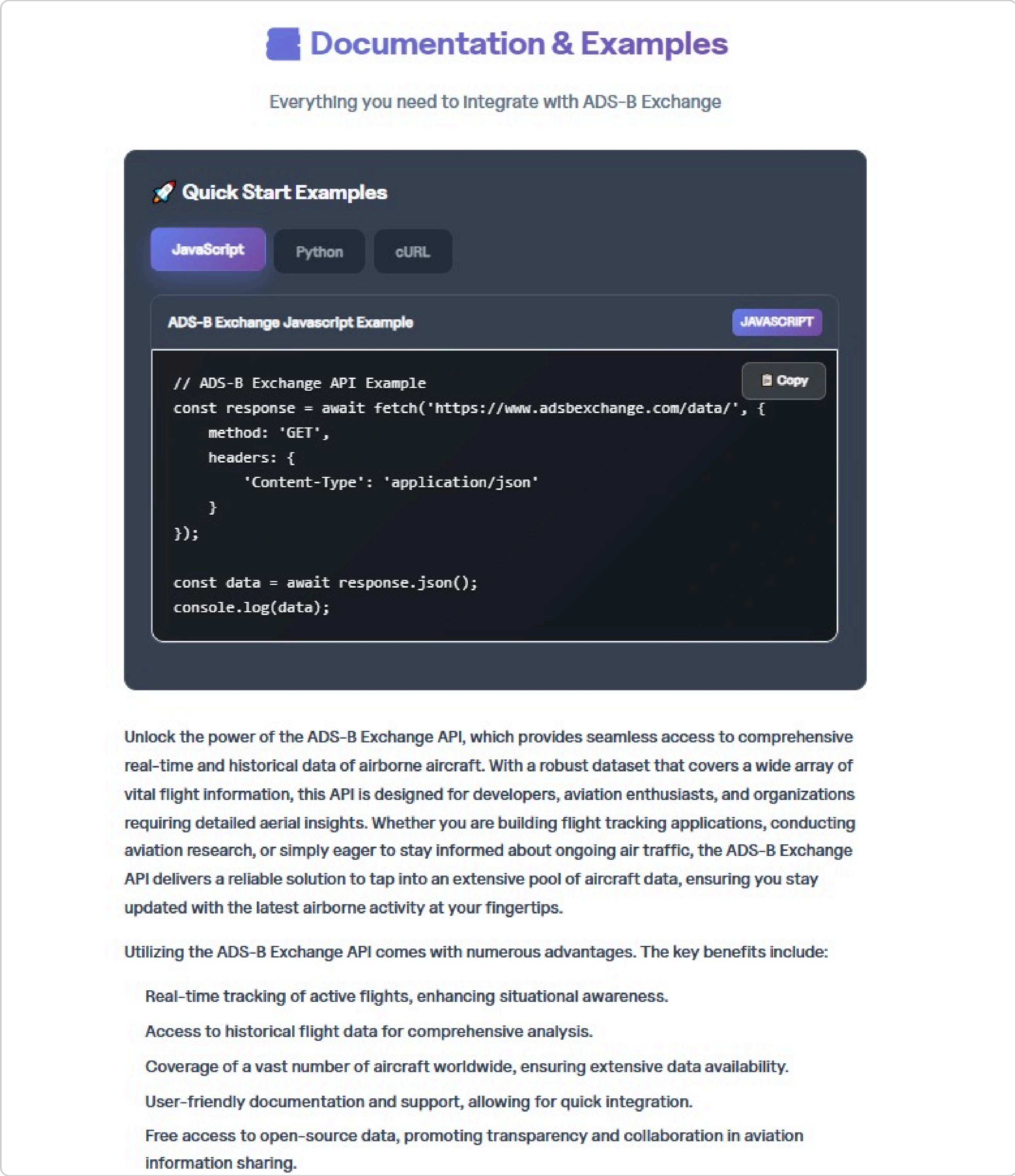


Figure 2: Live context panel showing weather, feed health, and active board state for tactical decision-making.

Access

Live link: https://jfk-ops-live-dashboard-vercel-mirro.vercel.app/